Product Identity

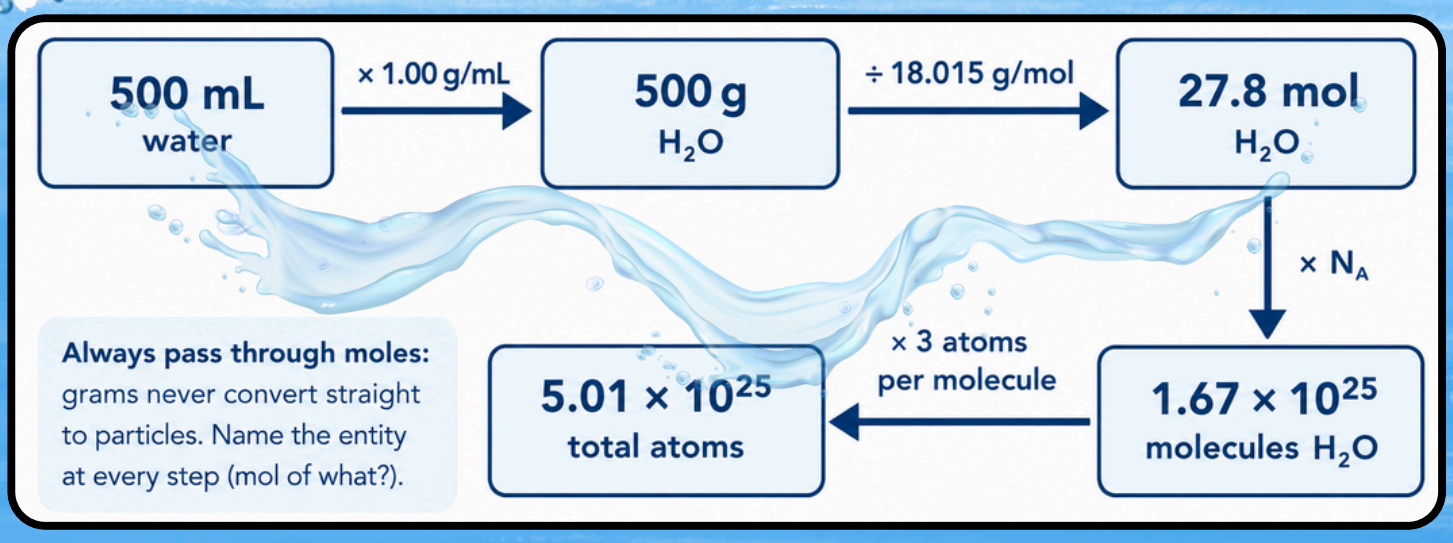


Product Name:
Bottled Drinking Water
Brand:
Summit
Product Type:
Drinking Water
Chemical Compound:
Water
Chemical formula:
H_2O
Package size:
500 ml.
Student:
Coervin James K. Ariola, Chemisty for Engineers
Purpose:
Water is issential for life to exist, it is widely used on laboratories, cooling systems, and industrial processes.

Base Quantitative Data

Quantity	Value	Unit	Source
Net Content	500	ml	Products' Label
Density of Water	1	g/ml	Assumed Value
Molar Mass of Water	18.015	g/ml	Calculated From Atomic Masses
Avogadro's Constant	$6.02214076 \times 10^{23}$	mol^{-1}	Exact SI Constant
Hydrogen Atomic Mass	1.008	g/mol	Standard Atomic Mass
Oxygen Atomic Mass	15.999	g/mol	Standard Atomic Mass

The Mole Map



Molar Mass Build-Up

Element	Number of Atoms	Atomic Mass	Contribution
Hydrogen	2	1.008	2.016 g/mol
Oxygen	1	15.999	15.999 g/mol
Total	3	-	18.015 g/mol

Mole Conversion Calculation

Molar Mass:
$M=2(1.008)+15.999$
$=18.015$ g/mol
Label Bridge:
500 ml * 1.00g/1 ml
=500 g water
Mass to Moles:
$n=500g/18.015g/mol$
27.8 mol H2O
Moles to Molecules:
$N=27.7546*6.02214076*10^{23}$
$1.67*10^{25}$ molecules
Percent Composition and Two-Route Check:

Hydrogen: %H=2.016/18.015*100 =11.19%

Oxygen: %O=15.999/18.015*100 =88.81%

Route A Using Percent Composition:

500 g*0.1119067=55.95 g H

Route B Using Mole Ratios:
Route A Using Percent Composition:
27.7546*2*1.008=55.95 g H

Atoms of One Element:

Hydrogen Atoms: $1.6711*10^{25}*2$
$3.34*10^{25}$ H Atoms

Oxygen Atoms: $1.6711*10^{25}*1$
$1.67*10^{25}$ O Atoms

Total Atoms: $3.34*10^{25}+1.67*10^{25}$
$5.01*10^{25}$ Atoms

Significant Figure Check:

Given Volume: 500 mL
Assumed Density: 1.00 g/mL (3 significant figures)
Molar Mass of Water: 18.015 g/mol
Mass of Water: 500 g
Moles of Water: 27.8 mol
Molecules of Water: 1.67*10^25 molecules
Total Atoms: 5.01*10^25atoms

Scale-Up:

Number of Bottles: 40
Water Mass: 20 kg.
Total Atoms: 2.01×10^{27}
Total Volume: 20 L

One-Number Takeaway:

Water can be measured in different ways, such as volume, mass, moles, molecules, and atoms. Using the density of water, its molar mass, and Avogadro's number, we can convert from one quantity to another. A single 500 mL bottle contains approximately 1.67*10^25water molecules and 5.01*10^25 total atoms, assuming a density of 1.00 g/mL. This shows how the mole connects everyday products to the incredibly large number of particles they contain.

Referrences:

- National Library of Medicine. (n.d.). Water. PubChem. https://pubchem.ncbi.nlm.nih.gov/compound/Water
- National Institute of Standards and Technology. (n.d.). Water. NIST Chemistry WebBook. https://webbook.nist.gov/cgi/cbook.cgi?Name=water